Smart Cane: Better Walking Experience For Blind People

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Abstract—**Today innovation is enhancing every day in various viewpoints so as to give adaptable and safe development to the general population. Outwardly disabled individuals discover troubles recognizing obstacles before them, amid strolling in the road, which makes it unsafe. The smart cane has a system that empower them to see and differentiate the obstacles. Moving with the assistance of a white cane is a slippery errand for the outwardly tested. The smart cane is equipped with infrared and ultrasonic sensor that helps in detecting the obstacles. At the base of the smart cane there is a water sensor which detects and dodges puddles. When it recognizes any obstacle ,it activates the sound system and the vibration motor. On detecting obstructions the sensor passes this information to the micro-controller. The micro-controller then procedures this information and computes if the object is sufficiently close. In the event that the object is not that close the circuit does nothing. In the event that the object is close the micro-controller sends a signal to sound a buzzer. GPS system provides the information about the current location in case of an emergency.**
Keywords-Infrared, Ultrasonic,Micro-controlller,GPS

I. INTRODUCTION

In a survey organized by World Health Organization (WHO, 2011), there are 285 billion people in the world are suffering from visual impairment and related problems of which 39 billion have no sight or blind and 246 billion suffering from low sightedness. There are around 15 million people with visual impairment in India and it is going to further increase the future years. A research study revealed that 2% of the population uses mobility aids In France.

Smart Cane device is an electronic travel device that fits on the top of the white cane. It is more improved version of white cane as it eliminates the drawbacks of white cane by detecting the hanging obstacles and knee above. Smart sticks record the information about the obstacles present in the way by using The location can be set by voice and the user can be further guided by the GPS to his/her distention. It uses ultrasonic sensors for detection and a micro-controller for controlling the system. . It also contains GPS which has a memory card used to store various locations. It works on rechargeable batteries that can be recharged using USB or an AC adaptor. The accuracy of the GPS signal is poor as it can be controlled within 5 meter radius only. However, this system is not made for indoor purposes due to poor GPS signals. A camera is fitted on the head of the person using it, which uses an algorithm to detect the obstacles ahead. Finally, the visually impaired person needs to be properly trained to use it effectively. This paper proposes a design of smart cane for the visually impaired people that helps them to navigate in the public places.

[2]Overcoming some of the disadvantages:

- To detect and speak aloud upward and downward stairs.
- Instead of incomprehensible sound, providing aid of earphones which give the speech warning message and avoid public embarrassment.
- Providing integrated vibration motor that helps in giving feedback and equipped in the hand of the smart stick to keep it user friendly.

II. LITERATURE SURVEY

1. Smart Blind Walking Stick: Vipul V. Nahar. In this paper, the circuitry consist of ultrasonic sensors, moisture sensors and also feedback system.The micro-controller PIC16F876 processes the inputs from the sensors and calculates the distance between the blind and obstacle and triggers the buzz sound if it is within desired distance. [4] They also implemented one more feature which helps the blind to find his/her stick by using wireless RF remote. The advantage of this system is that it is inexpensive and can be implemented by millions of blind people.
2. Effective Fast Response Smart Stick for Blind People: Ayat Nada. They design smart stick equipped with infrared, ultrasonic and moisture sensor. The selection process of all the sensors are based on factors such as cost, kind of obstacles,

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detection range and atmospheric conditions. The micro-controller used is 18F46K80. The micro-controller process the inputs from the sensors and activates the vibrating motor and invokes the warning message. The stick can detect the obstacles within 4 m, able to give ETAs and also the stick has low power consumption as well as light weight and flexible [2].

3. Ultrasonic Stick for Blind: Deepak Kumar. Uses stick equipped with ultrasonic, GSM, GPS, camera and a micro-controller. Here the micro-controller used is PIC18F45K22. The sensors detects any obstacles within 180 degree path and triggers the buzzer. GPS can be set by voice commands and system will guide the user to desired destination. 2 micro-controller is used, one for sensors and other for GPS navigation. There is also an additional feature like emergency button which will in case of emergency message the location of blind to all the saved numbers. [1]
4. Sound and Touch based Smart Cane: Better Walking Experience for Visually Challenged: Anjali Gopinath. They divided the smart cane into 3 parts heat and obstacle detection and Bluetooth module. The distance is measured using ultrasonic sensor and data is sent it to the Arduino module. Arduino then processes the data and gives desired output in the form of message and buzzer. The main feature of this paper is that, in this model smart cane is able to identify the obstacles of the basis of its temperature and informs the blind person about how hot is the obstacles is. [3]

III. SYSTEM DESCRIPTION

The laser sensors can’t give an accurate reading for the close obstacles(less than 15 cm) but the ultrasonic sensors work well for close obstacles. They are low in cost and their range is from 3 cm to 10 m. Their major advantage is that they are not affected by the atmospheric conditions.

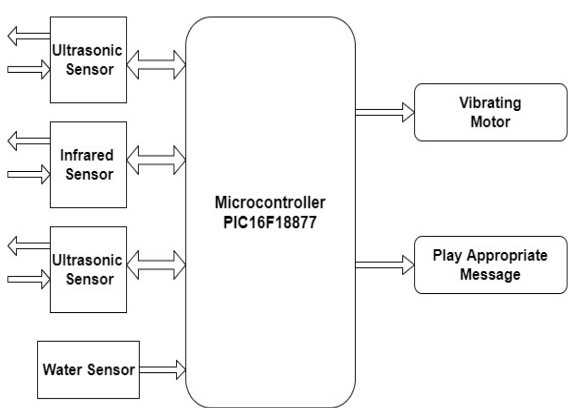

Fig.1. System Components of Smart Cane

The model uses two ultrasonic sensors, one at a height of 90 cm and other at 30 cm for detecting both upper and lower obstacles.

The model uses two ultrasonic sensors, one at a height of 90 cm and other at 30 cm for detecting both upper and lower obstacles. The ultrasonic sensors detect on the basis of time of flight and beam size. The vibration motor activates if the sensors detect any obstacle in 180º horizontal and 60º vertical.

Infrared Sensors work by transmitting and receiving pulse of IR light. Their range is from 20 cm to 150 cm. Their beam width is thinner than that of ultrasonic sensors. They are used for detection of upward/downward stairs.

Micro-controller is a SoC which is integrated on a circuit consist of processor, memory and I/O peripherals. The micro-controllers are used for controlling devices and products automatically such as medical devices, remote control toys and in this case smart cane. The micro-controller used is PIC16F18877. The micro-controller has 40 pin and a memory of 56KB, RAM of 4096 bytes and 256 bytes of EEPROM. It requires voltage ranging 1.8v to 5.5v and operated between -40 C and 125 C [10]. The micro-controller controls the embedded system with processor and memory support. The PIC is turned on when it runs the program into its memory and doesn’t have any operating system. The PIC is also has a regulator which regulates the voltage to +5V.

Vibration Motor used in smart stick are similar to dc vibration motors that are used in mobile phones. They require 1.3v-3v of a voltage supply and around 125 mA current. They can also be programmed to control their speed of vibration. Their speed is about 13500 , have a diameter of 0.5 cm and thickness of 2.5 mm [1].

When the micro-controller sends a message to the GSM modem, it processes that message and gets the location of the stick from GPS modem and transmits it to the GSM modem to respond to the sender [8]. The stick also has an emergency button which the user can press to send location through SMS to all the saved numbers in the micro-controller.

Water sensors are used in the smart sticks for the detection of water existence. Two wire probes are fitted at its bottom to detect water pits/puddles on the road. If these wire probes touch the water, the circuit is shorted and it starts the motor with a sound message : "There is water".

IV. WORKING OF THE PROPOSED SYSTEM

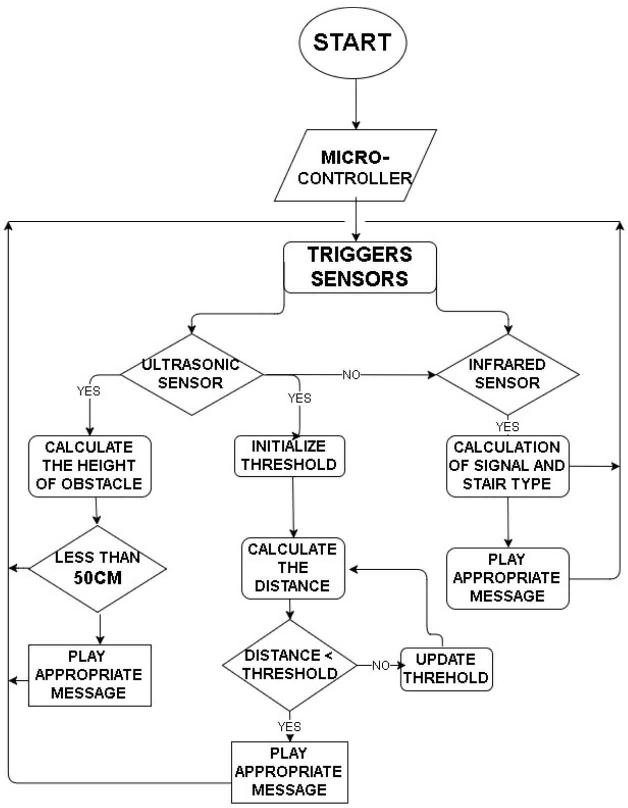

Fig.2. Flowchart of the proposed system

For obstacle detection, to cover all the objects around the stick the infrared sensor is mounted on top of the stick with correct angle, as if it is placed on the lower side it is able to detect only the lower objects. It is necessary to test all the probable areas to place the sensors

As shown in the above figure, when the micro-controller is started it produces 40 KHz wave which drives the ultrasonic emitter. After sending the wave, it then collects the received wave from the ultrasonic sensor and convert it into digital form. If the system received signals from ultrasonic sensors then it calculates the distance and if the IR is detected, the micro-controller calculates the average of the signals and invokes the appropriate message with different buzzer sound, if there is no signal detected, the control goes back to the initial stage. When ultrasonic signals are detected, the micro-controller then takes inputs from a pair of ultrasonic sensor that is, US2 and US1 sensors. US2 is detected the micro-controller calculates the height of the obstacles. If the height is within 50 cm, the system invokes no alarm but when it is more than 50 cm, the system invokes appropriate message to the blind person. When US1 sensor is detected, the micro-controller calculates the distance by taking time difference of transmitted and received signal. If the distance is less than threshold which in this case is 4m, the system invokes no message and alarm and control goes to update the threshold distance. If the distance is more than threshold distance, the micro-controller then outputs appropriate warning distance. At the end of the stick, there is moisture sensor which on detection give inputs wave to micro-controller and it invokes the puddle warning message. The stick is also equipped with GPS navigation is used to tell the path to the blind person.

V. RESULTS AND ANALYSIS

In this section the system was tested to make sure that its functions are working properly. The section analyses some of its modules. The ultrasonic, infrared and water sensors are tested individually. Given below is the table for ultrasonic sensors, which is used to examine how efficiently the sensor is working.

TABLE 1 : ULTRASONIC SENSOR VALUES

Distance(cm)	Calculated Value(mV)	Measured Value(mV)	Error
5	20	21	1 mV
15	45	46.5	1.5 mV
25	90	94.4	4.4 mV
30	150	148	2 mV
40	200	195.3	4.7 mV
50	250	245.7	4.3 mV
75	375	368	7 mV
100	500	488.4	11.6 mV
150	750	732	18 mV
200	1000	974.8	25.2 mV
250	1250	1221.1	28.9 mV
300	1500	1468.2	31.8 mV
350	1750	1712.9	37.1 mV
400	2000	1953.2	46.8 mV

The graph (shown below) is plotted between the calculated and measured value by using MATLAB functions.

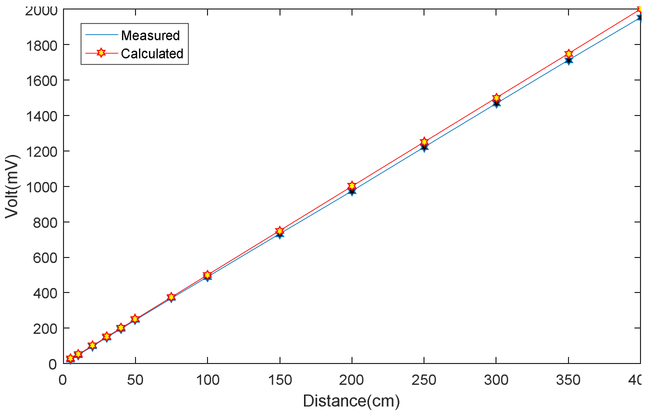

Fig.3. Graph between measured and calculated value

The output of the infrared sensor is dependent upon the distance and the size of the obstacle as well as material of the obstacle. To show this, the below graph shows the variation between the mean voltages and the distance between the blind and obstacles such as wood, plastic, steel, grass [7].

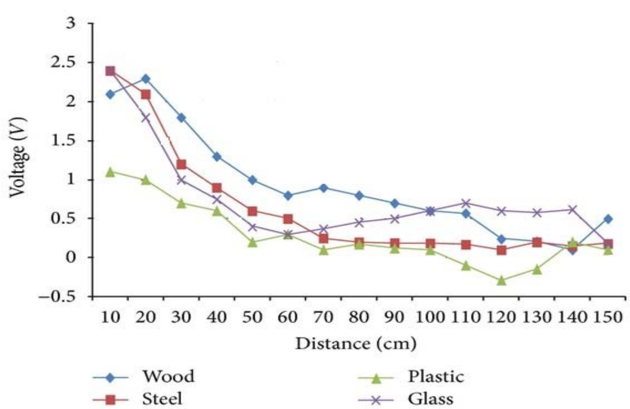

Fig.4. Graph between different obstacles for infrared sensors

Also testing of the voice message system been performed. Different voice message are given to the user according to the distance between the obstacle and the blind person, ultrasonic and infrared sensors also take part in this system [3].

Details from the water sensor is also obtained as mentioned below:

- It detects water only if depth is more than 0.5cm.
- The sensor is moisture sensitive so to stop the buzzer, sensor needs to wiped or kept dry.

TABLE 2: VOICE MESSAGE SYSTEM

S.no	Voice Message	Distance(m)
1	No voice message	>4
2	Object is between 3-4 m in front of blind person	3-4
3	Object is in front of blind person	<2
4	Collision alert	<0.75

VI. START-UPS ON SMART CANE

E-stick is an electronic interface which has been intended for the visually challenged individuals to make them walk more productively by giving the sign and voice message about the nearness of any obstacles on their way. The stick additionally empowers the visually impaired man to wind up distinctly autonomous and furthermore lead their existence with less support from others. A handful of companies are creating these kind of techs that will serve the needs of these frequently overlooked people.

1. The Dring Smart Cane: It is launched in January 2017 at CES. It is developed by a French company 'Fayet' [6]. The stick is equipped with sensors and GPS module which are placed in handle and it can send text messages, call to a set contacts and give alerts when button is pressed to cane has fallen.
2. IIT-G: Developed by 24 students from IIT Gandhinagar in collaboration with Blind People's Association (BPA) [10]. The cane is priced at about Rs.1000 and equipped with infrared and ultrasonic sensors. The stick has different tones of buzzer sounds depending upon the shape, size and distance between the person and obstacle.
3. IIT- DELHI: IIT- DELHI has developed the smart cane for visually impaired. They used ultrasonic systems used in radar which is better than normal ultrasonic sensors as it gives wide range and more accuracy [9].
   The stick also has object detection mechanism which detects the obstacle within 3m. The stick has 1.8m of detection indoors and 3m outdoors and consist of feedback system with gives messages on failure of any sensors or low battery warnings. The cane is height adjustable and costs around Rs.3500.

VII. CONCLUSION

The Smart Stick goes as an important stage for the future implications of all the more supporting gadgets for visually disabled to be safer. It is affordable and effective. It prompts user in identifying the stairs, water pits within 4m range. This paper analysed the existing technology and didn't provide any implementation results.

The proposed system offers low power consumption, reliable and portable solution for navigation. The feature of this system lies in the fact that it is low cost solution and can be implemented by millions of blind person worldwide.

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